CNNs vs Vision Transformers under Data Scarcity

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Abstract

Convolutional networks embed strong image-specific inductive biases: locality, weight sharing, and translation equivariance. Vision Transformers do not, and must learn equivalent structure from data alone. We ask whether these biases still confer a measurable advantage when training data is scarce, and how that advantage interacts with data augmentation. We train a small ResNet-style CNN and a parameter-matched ViT-Tiny of roughly $271{,}000$ parameters each on CIFAR-10 from scratch, using an identical optimisation recipe at six training-set fractions from 1% to 100% and three random seeds per condition. The CNN outperforms the ViT by between 7.7 and 10.9 percentage points at every fraction, and the ViT does not catch up even at the full training set. In our single-seed RandAugment ablation, augmentation benefits the CNN more than the ViT at every fraction and slightly degrades the ViT at 1%, contradicting the hypothesis that augmentation can substitute for missing inductive biases. The data-efficient training of Transformers reported in larger-scale work therefore appears to depend on a combination of augmentation, distillation, and longer training rather than on any single ingredient.

1 Introduction

Convolutional Neural Networks (CNNs) have dominated modern computer vision benchmarks since the rise of AlexNet (Krizhevsky et al., 2012), building on a longer lineage of convolutional models such as LeNet (LeCun et al., 1998). A central reason for their success is that their architecture hard-codes visual priors aligned with the spatial structure of images: locality, weight sharing, translation equivariance, and robustness induced by subsampling. The recent emergence of the Vision Transformer (ViT) (Dosovitskiy et al., 2021) has unsettled this consensus by showing that a model with substantially weaker image-specific inductive biases than CNNs can match or exceed CNN performance when pre-trained at large scale, especially on ImageNet-21k or JFT-300M. Yet the original ViT paper itself reported that, at ImageNet scale and without strong regularisation, ViTs lag behind comparable ResNets by a few percentage points. This raises a research question that matters far beyond benchmark leaderboards: when training data is scarce, does architectural inductive bias still confer a measurable advantage over a transformer with weaker image-specific priors, and how quickly does that advantage erode as more data becomes available?

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The question is important for two reasons. First, the data-scarce regime is the practical reality for the vast majority of practitioners outside large industrial labs: medical imaging, scientific applications, and most academic research operate on datasets several orders of magnitude smaller than JFT-300M. Choosing the wrong architecture in this regime can cost several percentage points of accuracy that no amount of hyperparameter tuning will recover. Second, the comparison sharpens a foundational debate in deep learning about whether strong inductive biases remain valuable, or whether sufficient data and compute eventually render them redundant.

To address this empirically, we train a small ResNet-style CNN and a ViT-Tiny on CIFAR-10 at six fractions spanning two orders of magnitude (1%, 5%, 10%, 25%, 50%, 100%), with three random seeds per condition. Both models are matched to within 10% of each other in parameter count and trained under an identical optimisation protocol (AdamW with cosine annealing), so that any performance gap can be attributed to architecture rather than capacity or compute. We additionally run an ablation in which RandAugment is applied during training, in order to isolate the contribution of data augmentation across the two inductive-bias regimes.

Our findings confirm that the CNN's architectural priors confer a substantial advantage under data scarcity, and that this advantage persists even when the full training set is used. The CNN outperforms the ViT by 8.5 percentage points at the 1% fraction, and the gap remains at 7.7 points at the full 100% fraction. Counter to one natural prediction of the inductive-bias hypothesis, RandAugment yields a smaller lift for the ViT than for the CNN at every data fraction, weakening the view that data augmentation alone can substitute for the inductive biases ViTs lack.

2 Background and Related Work

The architectural assumptions embodied by our CNN model are classically exemplified by LeCun et al. (1998). In their presentation of LeNet-5, LeCun et al. operationalised a set of convolutional priors that have shaped CNN design ever since: *locality*, since each hidden unit receives input only from a small spatial neighbourhood; *weight sharing*, since units in the same feature map apply the same learned filter across spatial locations; *translation equivariance*, since shifting the input produces a correspondingly shifted feature map; and subsampling-induced robustness to small shifts and distortions. The authors argued that these architectural constraints, combined with end-to-end gradient-based learning, reduce the need for hand-engineered feature extractors by shifting much of feature extraction into a trainable architecture operating directly on pixel images. More broadly, LeNet-5 illustrates how prior knowledge about the structure of two-dimensional visual signals can be encoded in the architecture itself: local receptive fields exploit spatial correlations, weight sharing reduces the number of free parameters, and subsampling decreases sensitivity to exact feature locations. This is precisely the property our experiments aim to measure. At very small data fractions, the CNN we train is a scaled-down inheritor of LeNet's design philosophy, and its accuracy at 1% of CIFAR-10 probes the value of these built-in priors under data scarcity.

Dosovitskiy et al. (2021) challenged the necessity of these convolutional priors by showing that a model with substantially weaker image-specific inductive biases than CNNs can match or surpass state-of-the-art CNNs when pre-trained at sufficient scale. Their Vision Transformer (ViT) splits an image into fixed-size patches, linearly embeds them, adds positional embeddings, and processes the resulting sequence with a standard Transformer encoder. Unlike CNNs, ViT does not bake locality, two-dimensional neighbourhood structure, or translation equivariance into every layer; these spatial relationships must largely be learned from data. The authors therefore found that, when trained on mid-sized datasets such as ImageNet alone, especially without strong regularisation, ViTs underperformed comparable ResNets. They attributed this to the fact that "Transformers lack some of the inductive biases inherent to CNNs, such as translation equivariance and locality, and therefore do not generalize well when trained on insufficient amounts of data." However, when pre-trained on much larger datasets such as ImageNet-21k or JFT-300M, ViTs became competitive with or superior to strong CNN baselines. Their experiments therefore established a scaling trade-off rather than a simple rejection of architectural priors: convolutional priors are especially valuable when data are scarce, while their advantage can be offset when enough data are available to learn the relevant visual structure directly. Our study is positioned as a controlled, small-scale analogue of this scaling argument. Rather than sweeping pre-training corpus size from millions to hundreds of millions of images, we sweep the available training set of CIFAR-10 from 1% to 100%, holding the rest of the experimental setup fixed, and ask whether the same qualitative crossover emerges in miniature.

Touvron et al. (2021) subsequently showed that the data-hungriness of ViTs is not entirely intrinsic. Their Data-efficient image Transformers (DeiT) attain competitive ImageNet accuracy without external pre-training, using ImageNet alone. This improvement comes from a carefully tuned training recipe, including strong augmentation and regularisation techniques such as RandAugment, Mixup, CutMix, stochastic depth, and repeated augmentation. For the strongest models, the recipe also includes a transformer-specific distillation token. The distillation token allows the transformer student to learn from a strong teacher, typically a CNN, through the self-attention mechanism. Read alongside Dosovitskiy et al. (2021), DeiT suggests that the disadvantage of weaker image-specific priors can be mitigated by stronger training procedures and, most explicitly, by distillation from a CNN teacher, which can transfer convolutional inductive bias to the transformer student. In our setting we do not study distillation; instead, our augmentation ablation isolates one component of this broader compensation mechanism. By applying RandAugment to both architectures at every data fraction, we test whether synthetic variation introduced by augmentation benefits the ViT more than the CNN, particularly in the low-data regime where the ViT's weaker built-in priors should be most costly.

Taken together, these three works define the conceptual axis along which we measure our two architectures. LeCun et al. exemplify the CNN priors of locality, weight sharing, equivariant feature maps, and subsampling-induced robustness; Dosovitskiy et al. identify the data regime in which such priors matter most; and Touvron et al. show that the disadvantage of weaker image-specific priors can be partly mitigated by stronger training recipes and CNN-teacher distillation. Our contribution is not to extend any of these results theoretically, but to test their joint prediction in a tightly controlled, capacity-matched setting on a single small dataset, where the effects of architecture, data scale, and augmentation can be cleanly disentangled.

3 Method

3.1 Problem framing

Our aim is to isolate the role of architectural inductive bias in image classification as a function of the amount of training data available. We therefore compare two networks that differ as sharply as possible along the inductive-bias axis but are matched on every other axis we can control. The first is a small ResNet-style CNN that embeds the standard convolutional priors of locality, weight sharing, and translation equivariance directly into every layer. The second is a ViT-Tiny variant that does not bake any of these priors into its layers and must learn the relevant spatial structure from the data. Both networks are trained from scratch (no external pre-training), on the same stratified subsets of CIFAR-10, with identical optimisation hyperparameters and almost identical parameter counts. Under these constraints, any systematic difference in test accuracy across data fractions is attributable to inductive bias rather than to capacity, compute, optimisation, or data exposure.

3.2 CNN architecture

We implemented a compact ResNet-style convolutional neural network for CIFAR-10 classification. The model receives 32×32 RGB images and begins with a 3×3 convolutional stem that maps the 3 input channels to 30 feature channels, followed by batch normalization and ReLU activation. The main body contains three residual blocks. Each residual block consists of two 3×3 convolutional layers, each followed by batch normalization, with ReLU activations and an additive residual shortcut. The first block preserves the spatial resolution and channel count. The second block increases the channel count from 30 to 60 and downsamples using stride 2. The third block increases the channel count from 60 to 120 and also downsamples using stride 2. After the residual blocks, we apply global average pooling to reduce each feature map to a single value, followed by a final linear classifier that outputs 10 class logits. The CNN has 271,480 parameters.

This architecture encodes several image-specific inductive biases. First, the use of small 3×3 convolutional filters encourages locality, because each layer processes nearby pixels and local spatial patterns. Second, convolutional weight sharing means that the same filters are applied across all spatial locations, allowing the model to reuse learned visual features throughout the image. Third, the convolutional structure provides translation equivariance: shifting an image feature tends to shift the corresponding activation rather than requiring the model to relearn the feature at every possible location. These biases make CNNs especially suitable for image data and are central to our comparison with Vision Transformers under limited labeled data.

3.3 ViT architecture

We use a compact ViT-Tiny architecture designed to match the CNN baseline as closely as possible in parameter count while preserving the standard Transformer-based formulation introduced by Dosovitskiy et al. (2021). The model operates directly on sequences of image patches rather than on convolutional feature maps. A 4×4 patch embedding layer, implemented as a strided convolution with kernel size and stride equal to 4, partitions each 32×32 CIFAR-10 image into an 8×8 grid of non-overlapping patches, producing a total of 64 patches per image. Each patch is projected into an 80-dimensional embedding space, yielding a sequence of shape $(64, 80)$.

A learnable class token is prepended to the patch sequence before processing. This token acts as a global aggregation vector whose final representation is used for classification. Because the Transformer architecture itself is permutation-invariant, learnable positional embeddings of shape $(65, 80)$ are added to the sequence to encode spatial information about patch order and location. Dropout with probability 0.1 is applied after positional encoding.

The encoder consists of four standard Transformer blocks operating in a pre-normalisation configuration. Each block contains a multi-head self-attention layer with four attention heads followed by a two-layer feed-forward MLP with GELU activation and an expansion ratio of three. Residual skip connections are applied around both the attention and MLP sublayers. After the final encoder block, the class-token representation is layer-normalised and passed through a linear classifier producing the ten CIFAR-10 class logits. The complete model contains 270,010 trainable parameters, within 1% of the CNN baseline's 271,480 parameters.

Unlike CNNs, Vision Transformers do not hard-code image-specific inductive biases such as locality, weight sharing, or translation equivariance into their layers. In a CNN, neighbouring pixels interact immediately through local convolutional kernels, and the same learned filters are reused across all spatial locations. This strongly constrains the hypothesis space and allows convolutional networks to exploit spatial regularities efficiently, especially when data are limited. By contrast, the ViT processes images as sequences of patch embeddings and relies on self-attention to learn spatial structure directly from the data. Although positional embeddings provide information about patch order, the architecture itself does not inherently assume that nearby pixels are more strongly related than distant ones, nor that visual features should be reusable across locations. Consequently, ViTs generally require larger datasets, stronger regularisation, or more sophisticated training procedures to achieve the same level of generalisation as CNNs in low-data regimes. Our experiments are specifically designed to measure the effect of removing these convolutional priors under controlled capacity and optimisation conditions.

4 Experiments

Dataset. All experiments are conducted on CIFAR-10 (Krizhevsky, 2009), which contains $60{,}000$ colour images of resolution 32×32 across ten balanced classes, split by the original authors into $50{,}000$ training and $10{,}000$ test images. We carve a stratified 10% slice off the training set to serve as a validation set, used exclusively for model selection. From the remaining $45{,}000$ images we subsample stratified training sets at six fractions: 1%, 5%, 10%, 25%, 50%, and 100%. The official test set is left untouched and used only for the final accuracy reported per run. Images are normalised channel-wise using the standard CIFAR-10 mean and standard deviation.

Training protocol. Every combination is trained from scratch using AdamW (Loshchilov and Hutter, 2019) with an initial learning rate of $1{\times}10^{-3}$ and weight decay of $1{\times}10^{-4}$, a cosine annealing learning-rate schedule over the full training horizon, and a mini-batch size of 128. We train for 50 epochs in all conditions. The loss is standard cross-entropy. All randomness is seeded at the start of each run, and the same seed controls both the data subsampling and the parameter initialisation, so that runs are fully reproducible from the model, fraction, and seed identifiers

Evaluation protocol. During training we evaluate validation loss and validation accuracy at the end of every epoch and keep the best checkpoint by validation accuracy. After the final epoch this best checkpoint is loaded and evaluated once on the held-out test set; the resulting top-1 accuracy is the metric we report for that run. For each cell we run three independent seeds and report the mean and standard deviation of the resulting test accuracies. Figures showing test accuracy versus data fraction display the mean as a line and ± 1 standard deviation as a shaded band.

Capacity and compute control. A central methodological concern in comparing a CNN and a ViT is to avoid attributing to architecture an effect that is in fact due to capacity or compute. We control for this on three axes. *Parameter count*: the CNN and ViT used here have $271{,}480$ and $270{,}010$ trainable parameters respectively, a difference of well under 1%. The ViT's embedding dimension and depth were deliberately reduced from the more common ViT-Tiny configuration to achieve this match. *Optimisation*: both architectures are trained with the identical optimiser (AdamW), learning rate, weight decay, schedule (cosine annealing), batch size, and number of epochs; no per-architecture hyperparameter tuning is performed. *Data*: at every fraction both architectures see exactly the same stratified training indices, validation indices, and test set, controlled by the shared seed. These controls substantially reduce capacity, data exposure, and optimisation differences between the two arms.

4.1 Results

Table 1 reports the mean and standard deviation of top-1 test accuracy across three random seeds for both architectures at each of the six data fractions; Figures 1–3 visualise the corresponding curves.

Table 1: Baseline test accuracy on CIFAR-10 in percent top-1, reported as mean $\pm$ sample standard deviation over three seeds.

	Data fraction					
Model	1%	5%	10%	25%	50%	100%
CNN	40.7 ± 1.0	54.5 ± 0.3	61.2 ± 0.3	68.9 ± 0.3	75.1 ± 0.5	80.3 ± 0.2
ViT	32.2 ± 1.2	44.8 ± 0.7	50.3 ± 0.8	58.5 ± 0.3	65.2 ± 0.4	72.6 ± 0.3

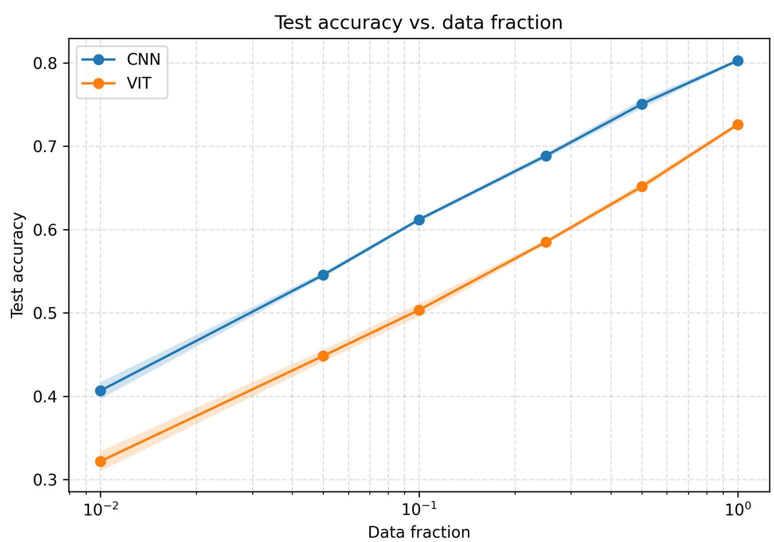

Figure 1: Test accuracy versus data fraction on CIFAR-10. Lines show the mean over three seeds; shaded bands show ± 1 sample standard deviation. The horizontal axis uses a logarithmic scale.

Figure 1 plots mean test accuracy against data fraction on a logarithmic horizontal axis, with ± 1 standard deviation displayed as a shaded band. Both architectures improve monotonically with more training data, rising from 40.7% (CNN) and 32.2% (ViT) at the 1% fraction to 80.3% and 72.6% respectively at the full 100% fraction. The CNN curve lies above the ViT curve at every fraction. The gap between the two means is 8.5 percentage points at 1%, increases to a maximum of 10.9 percentage points at 10%, and then decreases to 7.7 percentage points at 100%; the intermediate values at 5%, 25%, and 50% are 9.7, 10.4, and 9.9 percentage points respectively. The shaded bands are narrow throughout (per-cell standard deviations are at most 1.2 percentage points and most often below 0.5) and do not overlap at any fraction.

Figure 2 shows per-epoch training and validation loss for both architectures over the 50 epochs of training, for the 5% and 100% data fractions. At the 5% fraction (left panel) the CNN's training loss decreases to approximately zero within the first 25 epochs while its validation loss plateaus at roughly 1.5; the ViT's training loss decreases more gradually to approximately 0.3, and its validation loss reaches a minimum near epoch 10 and then rises again, ending above its initial value. At the 100% fraction (right panel) the train–val separations are smaller for both architectures: the CNN's training loss again approaches zero with validation loss settling near 1.1, and the ViT's training loss settles near 0.3 with validation loss reaching approximately 0.85 before slowly rising.

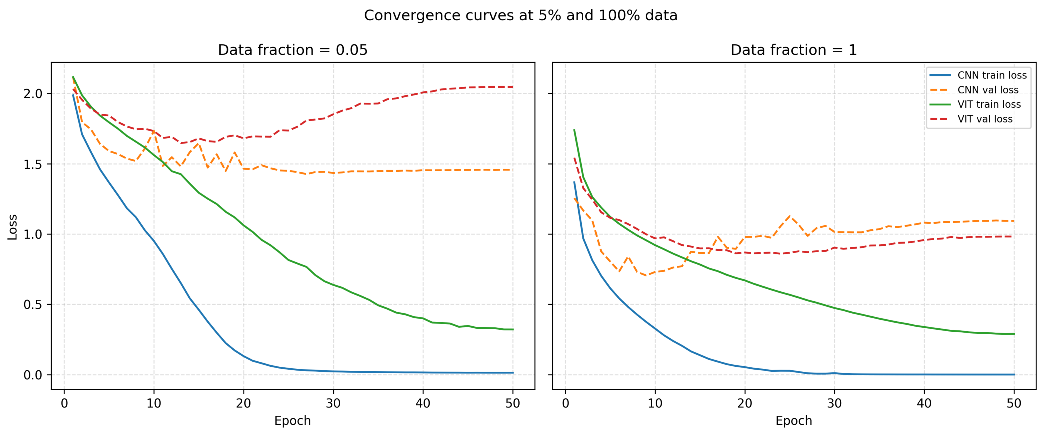

Figure 2: Training and validation loss per epoch at the 5% (left) and 100% (right) data fractions, for the CNN and the ViT.

Figure 3 reports the difference between final-epoch training accuracy and final-epoch validation accuracy, averaged across seeds, as a function of data fraction. For both architectures the gap decreases monotonically as the training fraction grows, from approximately 0.60 (CNN) and 0.52 (ViT) at the 1% fraction to 0.20 and 0.17 respectively at the 100% fraction. The CNN's gap is equal to or slightly larger than the ViT's at every fraction; the two curves nearly coincide between the 5% and 50% fractions and separate by approximately three percentage points at the extremes.

4.2 Ablations

Setup. To isolate the contribution of data augmentation, we re-trained both architectures at all six data fractions with RandAugment (Cubuk et al., 2020) applied to the training pipeline (`num_ops` $= 2$, magnitude $= 9$). All other hyperparameters were held identical to the baseline. Following the project specification, each augmented (model, fraction) combination was run with a single seed, in contrast to the three seeds used per baseline cell. We therefore compare the single augmented run against the three-seed baseline mean and report point estimates, rather than means with standard deviations, for the augmented condition.

Effect of RandAugment. Table 2 reports the augmented test accuracy alongside the change relative to the baseline mean ($\Delta = \text{aug} - \text{baseline mean}$), and Figure 4 visualises the corresponding deltas as a function of data fraction. With augmentation enabled, the CNN improves at every fraction: the gain is $+2.9$ percentage points at the 1% fraction, rises to a maximum of $+8.8$ percentage points at the 25% fraction, and remains positive but smaller at the 50% ($+7.5$) and 100% ($+6.4$) fractions. The ViT improves at five of the six fractions, by between $+4.7$ and $+6.4$ percentage points; at the 1% fraction, however, the ViT's test accuracy decreases by 0.8 percentage points. The ViT's gain is strictly smaller than the CNN's at every fraction; the difference $\Delta_{\text{ViT}} - \Delta_{\text{CNN}}$ ranges from -1.7 percentage points at the 100% fraction to -3.7 percentage points at the 1% fraction.

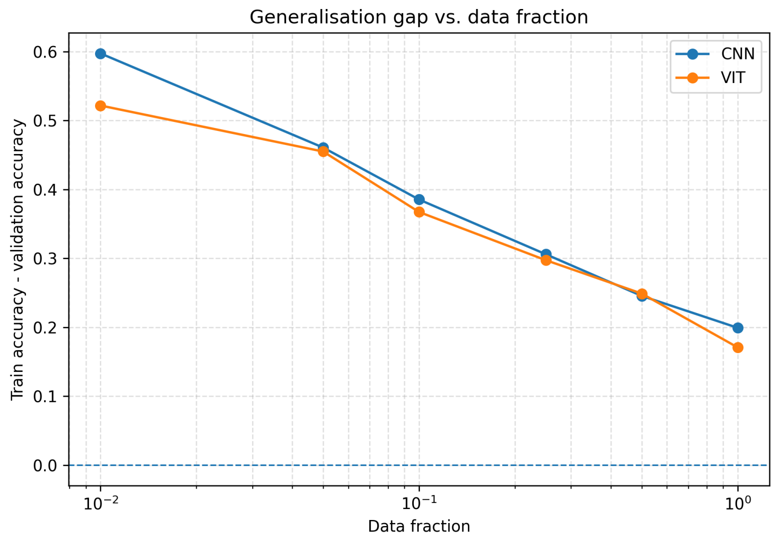

Figure 3: Generalisation gap (final-epoch training accuracy minus final-epoch validation accuracy, averaged across three seeds) as a function of data fraction. The horizontal axis uses a logarithmic scale.

Table 2: Effect of RandAugment on test accuracy (% top-1). Baseline values are the three-seed mean from Table 1; augmented values are from a single seed. Δ is the augmented accuracy minus the baseline mean, in percentage points.

	Data fraction					
	1%	5%	10%	25%	50%	100%
CNN + RandAugment	43.6	62.8	69.9	77.7	82.6	86.7
Δ (CNN)	+2.9	+8.3	+8.7	+8.8	+7.5	+6.4
ViT + RandAugment	31.4	50.6	55.5	64.9	70.8	77.3
Δ (ViT)	−0.8	+5.8	+5.2	+6.4	+5.7	+4.7
$\Delta_{\text{ViT}} - \Delta_{\text{CNN}}$	−3.7	−2.5	−3.5	−2.4	−1.8	−1.7

5 Discussion

Our baseline results confirm the first half of the inductive-bias hypothesis: when training data is scarce, the CNN's locality, weight sharing, and translation equivariance are decisive. At the 1% fraction the CNN reaches 40.7% test accuracy compared with the ViT's 32.2%, an 8.5 percentage-point gap that is many times larger than either model's seed-to-seed standard deviation. They do not, however, confirm the second half — that the gap should close as data grows. Within the range available on CIFAR-10, the gap actually widens before narrowing: it reaches a maximum of 10.9 percentage points at the 10% fraction and only falls back to 7.7 percentage points at the full 100% fraction. The ViT does not catch up at any fraction we tested. This is consistent with Dosovitskiy et al. (2021), who reported that ViTs only overtake CNNs once pre-training data reaches the JFT-300M regime; with at most $45{,}000$ images, our entire experimental range lies several orders of magnitude below their crossover, so the absence of a crossover is the expected outcome rather than a contradiction of their argument.

Figure 3 shows that the gap between training and validation accuracy decreases monotonically for both architectures as more data becomes available, from approximately 0.60 (CNN) and 0.52 (ViT) at the 1% fraction to 0.20 and 0.17 at the 100% fraction. Two features are worth noting. First, the CNN's generalisation gap is slightly larger than the ViT's at every fraction. This is not evidence that the ViT generalises "better" in absolute terms: the CNN's larger gap reflects the fact that it drives

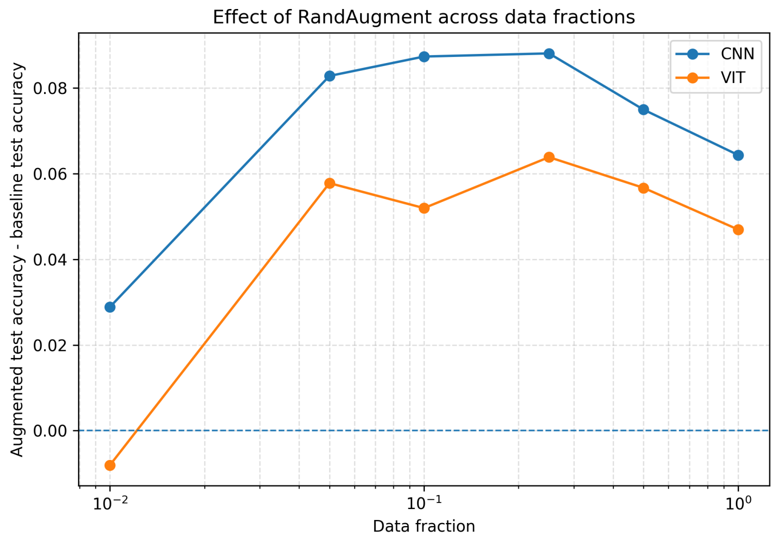

Figure 4: Change in test accuracy when RandAugment is applied during training, as a function of data fraction. Positive values indicate that augmentation improved test accuracy relative to the baseline (no-augmentation) condition. The horizontal axis uses a logarithmic scale; the dashed line at 0 separates conditions in which augmentation helps from conditions in which it hurts.

training accuracy close to 100% on the small training sets, while the ViT cannot fit the training set as tightly — a pattern visible in Figure 2, where the ViT's training loss remains around 0.3 throughout. Second, both gaps shrink at roughly the same rate, suggesting that additional data benefits the two architectures comparably in this regime; the architectures differ in their absolute accuracy, not in how quickly that accuracy improves with more data.

Our ablation runs return the most surprising finding of the study. The hypothesis we set out to test that data augmentation partially substitutes for missing architectural priors predicts that RandAugment should help the ViT more than the CNN, particularly in the low-data regime where the priors matter most. We observe the opposite. RandAugment improves the CNN by between 2.9 and 8.8 percentage points at every fraction, while improving the ViT by at most 6.4 percentage points, and at the 1% fraction it actively degrades the ViT by 0.8 percentage points. The augmented gap between CNN and ViT (Table 2) is therefore wider than the baseline gap at every fraction. This weakens the simple substitution hypothesis: RandAugment alone is not sufficient to compensate for what the ViT lacks. The most natural reconciliation with Touvron et al. (2021) is that their DeiT recipe couples RandAugment with several other ingredients, including Mixup, CutMix, stochastic depth, repeated augmentation, and crucially a transformer-specific distillation token that transfers convolutional inductive bias from a CNN teacher. Our results suggest that the burden of substituting for missing priors is borne by the full combination, not by augmentation in isolation. They also suggest that at very low data (1%, ~ 450 images), the additional variability that RandAugment introduces into the training distribution can be actively harmful for an architecture that lacks the spatial priors needed to anchor it.

Several aspects of our setup limit how widely our conclusions can be generalised. First, we evaluate on a single small benchmark; we cannot rule out that the picture differs at higher resolution or on more complex distributions. Second, both architectures are intentionally small; the inductive-bias trade-off may shift at the scale of ViT-Base or beyond. Third, our augmentation ablation uses a single seed per condition, so we cannot quantify variability for the augmented runs. Fourth, models are trained for only 50 epochs, and Figure 2 shows that the ViT's training loss is still decreasing at this horizon; a longer schedule might allow the ViT to make better use of the augmented data, particularly at the 1% fraction where it currently underperforms its baseline. Finally, we did not implement distillation, so our augmentation experiment does not isolate the question of whether *any* compensation mechanism can close the CNN–ViT gap at small scale.

6 Conclusion

We asked whether architectural inductive bias still confers a measurable advantage over a transformer with weaker image-specific priors when training data is scarce, and how that advantage interacts with data augmentation. On CIFAR-10, a parameter-matched comparison shows that the CNN's convolutional priors yield a 7.7–10.9 percentage-point accuracy advantage over a ViT-Tiny at every data fraction from 1% to 100%, and that RandAugment alone widens rather than narrows this gap. The most promising direction for genuinely closing it appears to be combining augmentation with a CNN-teacher distillation token, as in DeiT (Touvron et al., 2021), which we leave to future work.

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